

```

1 --s15_3 structural implementation
2
3 LIBRARY ieee;
4 USE ieee.std_logic_1164.ALL;
5 USE ieee.numeric_std.ALL;
6
7 ENTITY s15 IS
8     PORT (
9         clk : IN STD_LOGIC;
10        x : OUT STD_LOGIC_VECTOR(1 DOWNT0 0)
11    );
12END ENTITY;
13
14ARCHITECTURE behavioral OF s15 IS
15    SIGNAL q : STD_LOGIC_VECTOR(2 DOWNT0 0) := (OTHERS => '0');
16    SIGNAL a : STD_LOGIC;
17BEGIN
18
19    --flipflops
20    ff : PROCESS (clk)
21    BEGIN
22        IF rising_edge(clk) THEN
23            q(0) <= NOT q(0);
24            q(1) <= a;
25            q(2) <= q(1);
26        END IF;
27    END PROCESS;
28
29    -- combinational logic implementation
30    a <= ((NOT q(2)) AND q(1)) OR ((NOT q(1)) AND q(0)) OR (q(1) AND q(0));
31
32    -- output logic implementation
33    x <= q(1 DOWNT0 0);
34
35END ARCHITECTURE behavioral;

```